

“Hell is other people.” Despite Sartre’s claim, which suggests that the gaze of others is what blocks and anguishes you, the scientific literature shows that the gaze of others is one of the most relevant and information-rich cues that we extract from the social environment. From the first days of life, newborns show a preference for face-like stimuli and preferentially orient toward faces (Farroni et al., 2002; Johnson et al., 2015), and at the center of this attention are the eyes. Throughout this thesis, it will be analyzed the way in which gaze direction triggers the gaze cueing effect—one of the most studied phenomena in social attention (Friesen & Kingstone, 1998; McKay et al., 2021). Through the experiments carried out, the extent to which this effect, often considered automatic, can be modulated by the sense of agency will be investigated.

First, it is worth mentioning the morphological specificity of the human eye. Unlike other primates, the human eye has a white sclera, widely visible, surrounding a relatively small pupil and iris (Kobayashi & Kohshima, 1997). This configuration is not merely an anatomical detail, but has profound functional consequences: it makes gaze direction extremely easy to detect (Frischen et al., 2007), allowing the rapid and efficient extraction of social information through robust processing mechanisms (Langton et al., 2000). From an evolutionary perspective, two major explanations have been proposed for this development (see the discussion in Langton et al., 2000). On the one hand, the existence of specialized brain circuits for gaze processing, initially highlighted by Perrett and colleagues (Perrett et al., 1990, for animals; Haxby et al., 2000, for humans; Puce & Perrett, 2013), suggests that evolutionary pressures have dedicated specific neural resources to capturing information from the gaze of others. On the other hand, the structure of the eye itself seems to have evolved to be easily detected by the human visual system, turning gaze into a social signal that is impossible to ignore—as evidenced by infants’ early sensitivity to gaze direction (Batki et al., 2000). Moreover, this capacity is not exclusively human: gaze following has also been documented in other primates, which adjust their attention according to the gaze direction of conspecifics (Shepherd et al., 2006), suggesting a common evolutionary origin that predates the emergence of modern humans. Beyond the detection of gaze direction, direct eye contact—the actual meeting of gazes—constitutes a social signal with a special status: it signals the intention to interact, captures attention automatically, and activates neural networks involved in social cognition (Senju & Johnson, 2009). This direct contact is often the first step toward establishing joint attention, providing the context in which the deictic cue of gaze becomes truly meaningful. In this context, the gaze-cueing paradigm offers a privileged experimental window into how gaze direction orients our attention, and the factors that modulate it—including the sense of agency—become accessible to empirical investigation.

Beyond simple spatial orienting, gaze following fulfills a fundamental social function: it enables joint attention, that is, the ability to share with another person an attentional focus on the same object or event (Moore & Dunham, 1995). When we follow someone’s gaze, we not only shift our attention in the same direction, but we gain access to the other person’s attentional

focus, information that is essential for understanding what they think, want, or intend (Schilbach, 2015). This property of gaze, namely its ability to indicate a referent without naming it explicitly, has direct consequences for human communication. In dialogue, for example, sentences with an implied subject (such as “It is beautiful”) become intelligible precisely because the speaker’s gaze direction indicates the object to which they refer. Gaze thus functions as a spontaneous deictic cue, that is, a signal that directs someone’s attention to an object, location, or person without naming it explicitly, anchoring language in the immediate perceptual context and facilitating intersubjective understanding. This function of gaze as a deictic cue is considered an essential precursor to language acquisition, enabling the establishment of common reference in early interactions (Tomasello, 1995; Tomasello et al., 2005). This capacity was conceptualized by Baron-Cohen (1995) as being supported by a specialized cognitive module, namely the Eye Direction Detector (EDD), which functions as a precursor to theory of mind. In this sense, gaze following is not merely an attentional orienting mechanism, but a gateway to complex social cognition: detecting gaze direction enables the attribution of intentions and mental states to others, processes that underlie theory of mind (Baron-Cohen, 1995; Baron-Cohen et al., 1997).

Moreover, recent research emphasizes that purely observational paradigms, in which the participant passively views a face on a screen, do not capture the full complexity of the social processing of gaze. On the contrary, a “second-person neuroscience” perspective has emerged, arguing that understanding the behavioral and neural mechanisms of social gaze requires interactive approaches, in which the participant is actively engaged in generating or exchanging social cues (Schilbach et al., 2013; Schilbach, 2016). This transition is visible even in the evolution of research on the gaze-cueing effect, which has begun to be extended to paradigms involving real-time interactions and oculomotor measurements (Pfeiffer et al., 2013). It is within this theoretical framework that the central question of the present work arises: if simply observing a gaze triggers robust cueing effects, what happens when the participant is themselves the agent producing the cue? The shift from a passive observer to an active agent in gaze processing therefore constitutes the central stake of this thesis. Of course, the experimental paradigm used here, an adaptation of the Posner task (Posner, 1980), which we will illustrate shortly, does not involve face-to-face interactions or eye-tracking; nevertheless, by systematically manipulating the contingency between the participant’s action and the change in gaze direction, it represents a first controlled step toward investigating the role of the sense of agency in social attentional orienting.

All experiments in this thesis are based on the Posner paradigm, a classic paradigm in the study of attentional orienting, that later became fundamental for research on the gaze-cueing effect as well. Starting from the original article by Posner (1980), various versions have been developed, but they all preserve a basic structure: a central or peripheral cue is presented, often an arrow, indicating the probable location of an upcoming target. The difference in reaction times

is then calculated between valid trials, in which the cue correctly indicated the location of the upcoming target, and invalid trials, in which the cue indicated the wrong location. This difference represents the attentional benefit conferred by cue-based orienting. In the late 1990s, three research groups independently adapted this paradigm, replacing the arrow with a social cue: a human face whose gaze was directed to the left or to the right (Friesen & Kingstone, 1998; Driver et al., 1999; Langton & Bruce, 1999). The first study (Friesen & Kingstone, 1998) used a stylized version of a human face and demonstrated that gaze triggers reflexive, automatic orienting, independent of the strategic utility of the cue. The next two studies used realistic faces and highlighted the specific role of the eyes, showing that the cueing effect is due to gaze and not to head orientation (Driver et al., 1999; Langton & Bruce, 1999). This distinction has ecological relevance beyond the laboratory: in sports, for example, the effect of deceptive feints exploits precisely the dissociation between body (and head) posture and the actual direction of gaze, in order to mislead the opponent (Kunde et al., 2011; Meyer et al., 2022). In the same vein, Langton and Bruce (1999) demonstrated that gaze and arrows engage partially distinct attentional mechanisms: gaze is not merely a more “ecological” arrow, but a special social cue.

The seemingly minor modification of replacing an arrow with a gaze thus opened an entire field of research, demonstrating that attentional orienting is not governed exclusively by arbitrary symbols, but also by social signals with deep ecological relevance. At the same time, the two categories of cues differ in a fundamental respect: while arrows orient attention predominantly when they are predictive, gaze produces a robust effect even when the participant knows that its direction does not predict the target location (Friesen & Kingstone, 1998). This resistance to non-predictiveness is one of the defining features of the gaze-cueing effect and suggests that it engages attentional mechanisms distinct from those involved in symbolic cues—an aspect that will be examined in detail in what follows.

To better understand the nature of this effect and the mechanisms that govern it, it is useful to first detail the typical structure of a gaze-cueing experiment. This same structure underlies the three experiments on which the present work is based. It is as follows: the participant is seated in front of a screen and fixates a central point. In the center of the screen, a human face (schematic or photographic) appears, with its gaze initially directed straight ahead. After a short interval (usually between 300 and 1000 ms), the face’s gaze suddenly shifts to the left or to the right. At a variable interval after the onset of the cue (stimulus onset asynchrony, SOA)—that is, the interval between the onset of the cue and the onset of the target—a peripheral target appears. This target can be, for example, a letter, an asterisk, or a circle, but in the case of the present experiments, it is a square of a certain color. This target appears either in the direction indicated by the gaze (congruent trial) or in the opposite direction (incongruent trial). The participant’s task is to detect or discriminate the target as quickly as possible, usually by pressing a button. Both static faces (photographs with averted gaze) and dynamic faces (animated

sequences in which the eyes move) produce the cueing effect. In the static version, the face appears directly with averted gaze; in the dynamic version, the eyes move in real time. Both versions produce the cueing effect, and the differences in magnitude and mechanism will be discussed later. The gaze-cueing effect (Gaze Cueing Effect, GCE) is operationally defined as the difference between the mean reaction times recorded in incongruent trials and those recorded in congruent trials: $GCE = RT(\text{incongruent}) - RT(\text{congruent})$, expressed in milliseconds. A positive GCE indicates that participants responded faster to targets appearing in the direction of gaze, confirming that the social cue facilitated attentional orienting. The typical magnitude of this effect lies between 10 and 20 ms, and it has been replicated in hundreds of studies, being considered one of the most robust phenomena in the social attention literature (Frischen et al., 2007; McKay et al., 2021).

One of the particularly relevant aspects of the gaze-cueing effect is its profoundly automatic nature. Eloquent evidence in this regard comes from studies that manipulated the interval between cue and target (SOA). For example, one of the first experimental investigations of this effect, Friesen and Kingstone (1998), showed that although the effect is significant at SOAs of 105, 300, and 600 ms, it disappears completely at an SOA of 1005 ms. The fact that the effect is present at very short SOAs but disappears after approximately one second corresponds to the typical temporal profile of exogenous orienting: rapid in onset but transient, dissipating in less than a second (Müller & Rabbitt, 1989). We will define exogenous orienting as the orienting of attention triggered involuntarily by an external stimulus, independent of the observer's intentions. A familiar example is a sudden movement occurring in the environment, as when someone tries to startle us—a phenomenon we detect without necessarily being consciously aware of it. In contrast, endogenous orienting is slower, voluntary, guided by symbolic cues and by the observer's expectations (Posner, 1980).

Another important result of the study by Friesen and Kingstone (1998) is that the effect proved to be purely facilitatory: reaction times were faster in trials in which the gaze indicated the correct location compared to a neutral condition, without any significant slowing in trials in which the gaze indicated the wrong location. In other words, gaze produces an attentional benefit without costs, a pattern that differentiates reflexive orienting from strategic orienting.

Moreover, the effect persists under conditions that, theoretically, should abolish it if it relied on strategic, controlled processes. For example, gaze-triggered orienting remains robust even when the cue is presented very briefly and then masked, preventing elaborate conscious processing (Sato et al., 2007). It is worth noting that Sato and colleagues (2007) demonstrated the same automatic orienting effect for cues provided by head direction, suggesting that not only the eyes but also head posture can trigger attentional shifts without awareness. The effect is also maintained when gaze direction does not predict the target location better than chance (Galfano

et al., 2012) and even when the cue is counterpredictive, that is, when it systematically indicates the opposite location of the target (Kuhn & Kingstone, 2009; Tipples, 2008). Moreover, the effect becomes even stronger when attentional resources are depleted by perceptual distraction and increased task demands, making nonpredictive cues harder to ignore (Gregory & Jackson, 2020). This remarkable robustness is consistent with the idea that gaze-triggered orienting is a fundamental mechanism, deeply rooted in human cognitive architecture, as also suggested by its presence at very early ages (Batki et al., 2000; Reid & Striano, 2005).

Regarding the type of response, namely what participants had to observe, no change in the magnitude of the cueing effect was observed as a function of the type of response required, namely detection, localization, or identification, which confirms that gaze-triggered orienting is a basic mechanism, insensitive to the specific demands of the task (Friesen & Kingstone, 1998).

The automatic nature of the effect is also explained by neural evidence. More specifically, neuroimaging studies have implicated the superior temporal sulcus (STS) in coding gaze direction (Hoffman & Haxby, 2000; Pelphrey et al., 2003), the temporo-parietal junction (TPJ) in integrating social signals and signals related to the sense of agency (Schilbach et al., 2012), and fronto-parietal networks in implementing attentional shifts (Callejas et al., 2004). These multiple lines of evidence, which have focused on different levels of stimulus presentation, participant task, cue characteristics, and more, have illustrated the gaze cueing effect as a deeply rooted, rapid, transient attentional mechanism largely impermeable to voluntary control. This automatic aspect of the effect is the element studied through the possibility that a higher-order factor, namely the sense of agency, may modulate this effect.

It is important to note that this relationship is bidirectional: just as gaze influences information processing, the characteristics of the social context—including the nature of the relationship between the observer and the person gazing—also modulate, in turn, the magnitude of the gaze-triggered orienting effect (Dalmaso et al., 2020). More recently, computational models of evidence accumulation have suggested that the speeding of responses to gazed-at targets might reflect an increase in the efficiency of information processing at the indicated location, rather than a simple facilitation of attentional orienting (Alister et al, 2024). However, it remains an open question whether this processing boost is sustained over time or whether it dissipates rapidly after the initial orienting. This distinction is relevant for the present study: if the sense of agency over the gaze modifies not only attentional orienting but also the depth of information processing at the indicated location, then the modulation of the gaze-cueing effect by agency might reflect a strategic redistribution of cognitive resources, rather than merely a simple inhibition of an automatic reflex. Nevertheless, although robust, the gaze-cueing effect is not insensitive to context. A vast literature has demonstrated that its magnitude is modulated by a multitude of factors, ranging from the characteristics of the face providing the cue (emotional expression,

perceived social status) to traits of the observer (anxiety, gender, age) and contextual variables (group membership, trust history) (for a review, see Dalmasso et al., 2020; McKay et al., 2021). In the first instance, the characteristics of the face providing the cue play a critical role: for example, negative emotional expressions, such as fear, reliably amplify the effect, although the results are not always consistent (Bayless et al., 2011; Hietanen & Leppänen, 2003). Secondly, observer traits shape the attentional response. For instance, women tend to show a stronger gaze-cueing effect than men (Bayliss et al., 2005), and older age is associated with a decline in the reflexive components of gaze-triggered orienting (Slessor et al., 2010; Kuhn et al., 2015). Thirdly, the social and interactional context, for example the perceived social status of the face or the history of cooperation, also modifies the magnitude of the effect. This permeability to social and cognitive factors underscores the fact that gaze-triggered orienting, although it has a reflexive component, is deeply integrated into higher-level social cognition processes. One aspect that has received surprisingly little attention in the literature is how the sense of agency—that is, the subjective experience of being the cause of an action and of its consequences (Haggard & Tsakiris, 2009)—modulates this effect. Therefore, if the GCE is permeable to modulation, and agency is an internal factor generated by action, then investigating this relationship becomes the next natural step.

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